Statistics of Financial Markets

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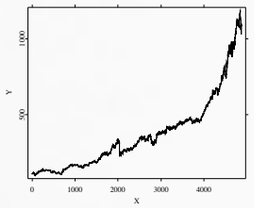
Chapter 3:
Concepts of Probability Theory



Random Variables

Variables like a stock price or a maximum temperature of a day are not completely predictable. However one knows which values are likely in the next observation period.

Let $X \in \mathbb{R}$ be such a random variable (rv), e.g. the stock price.

The **likeliness** of certain values of X is measured by the **probability of the events**

$$\{a < X < b\} \text{ or } \{X \leq b\}\,.$$

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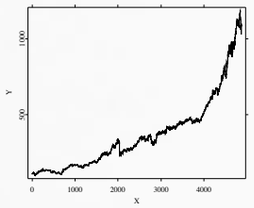


Distribution $\mathscr{L}(X)$ of X

The set of all probabilities

$$\mathrm{P}(a < X < b)\ , \qquad -\infty < a < b < \infty$$

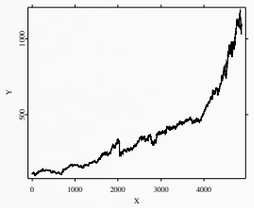
is called the distribution of X.

The probability density function (pdf) The pdf $p(x)$ of $\mathscr{L}(X)$ is defined by:

$$\mathrm{P}(a < X < b) = \int_a^b p(x)dx$$

For small h , $\qquad \mathrm{P}(x - h < X < x + h) \approx 2h \cdot p(x)$

hence $p(x)$ is a measure of the chance that X takes values near x.



Special Distributions

Normal Distribution, $\mathrm{N}(\mu, \sigma^2)$

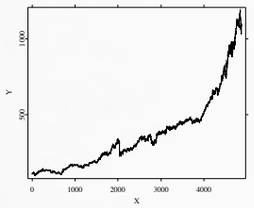
$$\varphi_{\mu,\sigma^2}(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{ -\frac{(x-\mu)^2}{2\sigma^2} \right\}$$

If $\mu = 0$ and $\sigma^2 = 1$, we have the standard normal distribution.

$$\varphi(x) = \varphi_{0,1}(x) = \frac{1}{\sqrt{2\pi}} \exp\left\{ \frac{-x^2}{2} \right\}$$

It is easy to check that

$$\varphi_{\mu,\sigma^2}(x) = \frac{1}{\sigma}\varphi\left(\frac{x-\mu}{\sigma} \right)$$

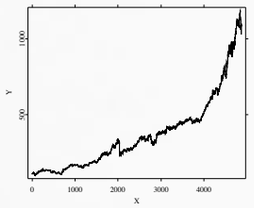


The cdf of a standard normal variable X:

$$\Phi(x) = \int_{-\infty}^{x} \varphi(y)dy$$

If $X \sim \mathrm{N}(\mu, \sigma^2)$, $(X - \mu)/\sigma$ is standard normally distributed

$$F(x) = \mathrm{P}(X \leq x) = \mathrm{P}\left(\frac{X-\mu}{\sigma} \leq \frac{x-\mu}{\sigma}\right) = \Phi\left(\frac{x-\mu}{\sigma}\right)$$



Log-Normal distribution

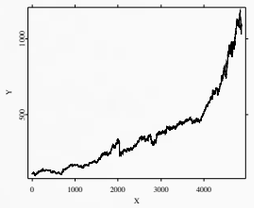
X is log-normally distributed, if $Y = \log(X) \sim \mathrm{N}(\mu, \sigma^2)$.

Distribution function of the log-normal:

$$F(x) = \mathrm{P}(X \leq x) = \mathrm{P}(\log X \leq \log x) = \Phi\left(\frac{\log x - \mu}{\sigma}\right), \quad x > 0$$

Log-Normal density function:

$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \frac{1}{x} \exp\left\{-\frac{(\log x - \mu)^2}{2\sigma^2}\right\} = \frac{1}{\sigma x} \varphi\left(\frac{\log x - \mu}{\sigma}\right), \quad x > 0$$



Discrete distribution

Distribution of a rv's which takes finitely many values is completely determined through

$$\mathrm{P}(X = x_k)\ , \quad k = 1,\ldots,n$$

Binomial Distribution

If $Y_1, \ldots, Y_n$ are independent Bernoulli distributed rv taking values in $\{0,1\}$ with probabilities:

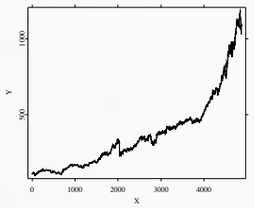
$$p = \mathrm{P}(Y_k = 1)\ ,\ 1 - p = \mathrm{P}(Y_k = 0)\ , \quad k = 1,\ldots,n\,.$$

the sum $X = \sum_{k=1}^{n} Y_k$ is binomial distributed $B(n,p)$ with parameters n, p:

$$\mathrm{P}(X = m) = \binom{n}{m} p^m (1-p)^{n-m}\,, \quad m = 0,\ldots,n\,.$$

SFEBinomial

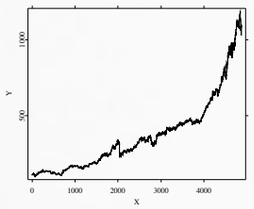
Jacob Bernoulli on BBI:



Central Limit Theorem (CLT)

If n is large enough a rv $Z \sim B(n, p)$ can be approximated by a rv $X \sim \mathrm{N}\{np, np(1-p)\}$ in the sense that

$$\mathrm{P}(a < Z < b) \approx \mathrm{P}(a < X < b)$$



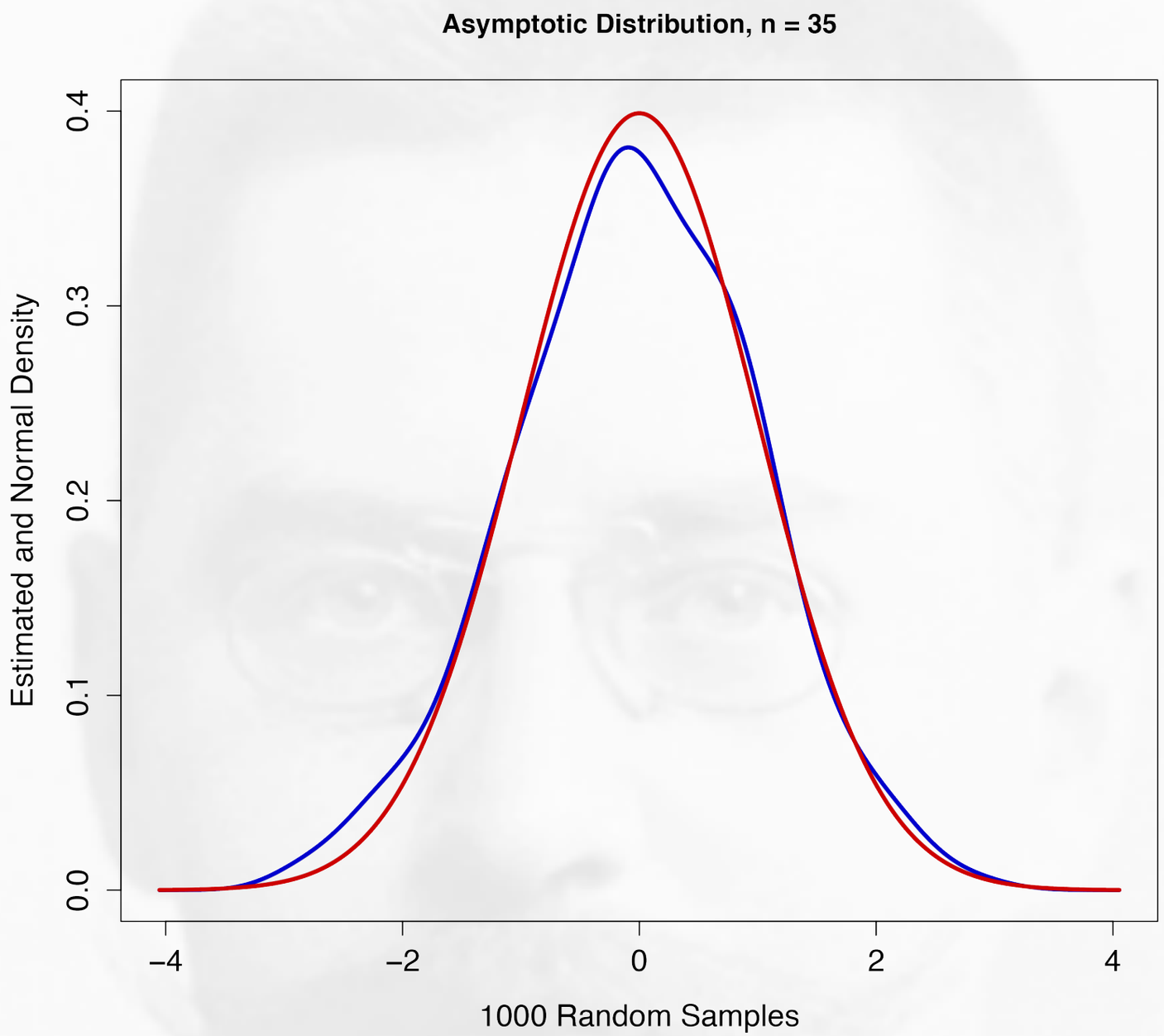
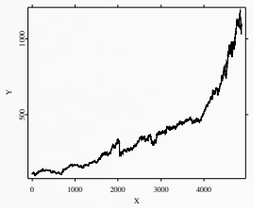


Figure 8: CLT estimate for Bernoulli distributed random variables (blue) together with the standard normal distribution (red).

SFEclt



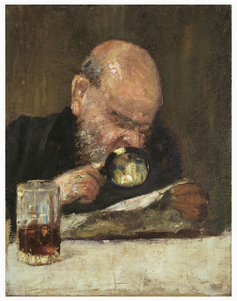
Expected value and variance

The expected value E[X] measures where the average value of X can be found.

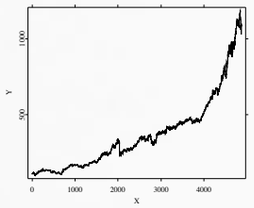
$$\mathrm{E}[X] = \int_{-\infty}^{\infty} x p(x) dx \quad \text{or} \quad \mathrm{E}[X] = \sum x_i \, \mathrm{P}[X = x_i]$$

If $X_1, \ldots, X_n$ are iid random variables one may estimate E[X] by

$$\hat{\mu} = n^{-1} \sum_{t=1}^{n} X_t$$



More details in the SFE book



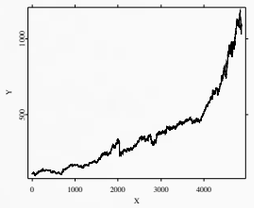
The **variance** $\mathrm{Var}(X)$ is a measure of the variation of X around the expected value E[X]

$$\mathrm{Var}(X) = \mathrm{E}\left[(X - \mathrm{E}[X])^2\right] = \int_{-\infty}^{\infty} (x - \mathrm{E}[X])^2 p(x) dx$$

The **standard deviation** $\sigma(X) = \sqrt{\mathrm{Var}(X)}$

One may estimate $\mathrm{Var}(X)$ by

$$\hat{\sigma}^2 = n^{-1} \sum_{t=1}^{n} (X_t - \hat{\mu})^2$$



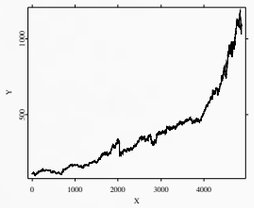
For a Log-normal rv $\log(X) \sim \mathrm{N}(\mu, \sigma^2)$:

$$\mathrm{E}[X] = \exp\left(\mu + \frac{1}{2}\sigma^2\right)$$

$$Var(X) = \exp\,(2\mu + \sigma^2)(e^{\sigma^2} - 1)$$

For $X \sim B(n, p)$:

$$\mathrm{E}\,[X] = np \text{ and } \mathrm{Var}(X) = np(1-p)$$



Skewness and Kurtosis

The **skewness** of a random variable X is:

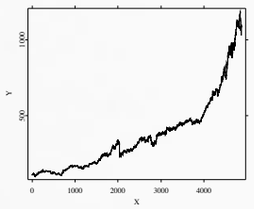
$$S(X) = \frac{\mathrm{E}\left[(X-\mu)^3\right]}{\sigma^3}$$

If $X \sim \mathrm{N}(\mu, \sigma^2)$ it follows $S(X) = 0$.

An estimator of the skewness from an iid sample is given by:

$$\hat{S}(X) = \frac{n^{-1}\sum_{t=1}^{n}(X_t - \hat{\mu})^3}{\hat{\sigma}^3}$$

(3.1)



The **kurtosis** is of a random variable X is:

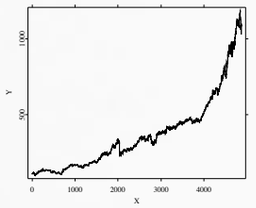
$$\text{Kurt}(X) = \frac{\text{E}[(X-\mu)^4]}{\sigma^4}$$

If $X \sim \text{N}(\mu, \sigma^2)$, $\text{Kurt}(X) = 3$

An estimator of the kurtosis from an iid sample is given by:

$$\widehat{\text{Kurt}}\,(X) = \frac{n^{-1}\sum_{t=1}^{n}(X_t - \hat{\mu})^4}{\hat{\sigma}^4} \tag{3.2}$$

A high kurtosis (>3) means that extreme values of X occur with high probability.

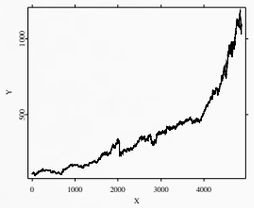


Random Vectors, Dependence, Correlation

The introduced concepts can be extended to several random variables. If we are interested in several variables $X_1, \ldots, X_n$ (e.g. several asset values) and their relative dependence, we study the random vector $(X_1, \ldots, X_n) \in \mathbb{R}^n$.

The distribution is

$$\mathrm{P}(a_1 \leq X_1 \leq b_1, \ldots, a_n \leq X_n \leq b_n) = \int_{a_n}^{b_n} \ldots \int_{a_1}^{b_1} p(x_1, \ldots, x_n)\, dx_1 \ldots dx_n$$



Independence

If X_1 and X_2 are independent, then

$$\mathrm{P}(a_1 \leq X_1 \leq b_1, a_2 \leq X_2 \leq b_2) = \mathrm{P}(a_1 \leq X_1 \leq b_1) \cdot \mathrm{P}(a_2 \leq X_2 \leq b_2)$$

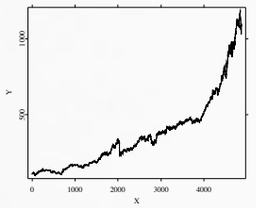
and $$p(x_1, x_2) = p_1(x_1)p_2(x_2)$$

If (X_1, X_2) are jointly normally distributed, the dependence may be characterized by the **covariance**

$$\mathrm{Cov}(X_1, X_2) = \mathrm{E}\{(X_1 - \mathrm{E}[X_1])(X_2 - \mathrm{E}[X_2])\}$$

or the **correlation**

$$\mathrm{Corr}(X_1, X_2) = \frac{\mathrm{Cov}(X_1, X_2)}{\sigma(X_1) \cdot \sigma(X_2)}$$



If $X_1, \ldots, X_n$ are independent

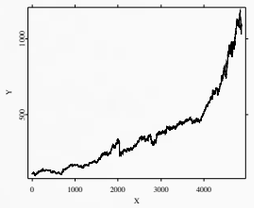
$$\mathrm{Cov}(X_i, X_j) = 0 \quad \text{for } i \neq j$$

$$\mathrm{Var}\left(\sum_{j=1}^{n} X_j\right) = \sum_{j=1}^{n} \mathrm{Var}(X_j)$$

If $X_1, \ldots, X_n$ are independent and have identical distributions

$$\mathrm{P}(a < X_i < b) = \mathrm{P}(a < X_j < b) \quad \text{for all } i, j$$

We say that $\{X_i\}_{i=1}^{n}$ are **independent identically distributed (i.i.d.)**



Conditional Probabilities

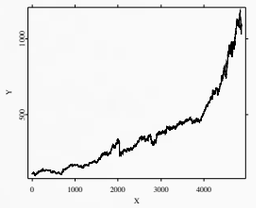
The **conditional probability** that Y takes values between a and b given that X takes values between x and $x + \Delta x$ is defined by

$$\mathrm{P}(a \leq Y \leq b \,|\, x \leq X \leq x + \Delta x) = \frac{\mathrm{P}(a \leq Y \leq b\,, x \leq X \leq x + \Delta x)}{\mathrm{P}(x \leq X \leq x + \Delta x)}\,, \tag{3.3}$$

The conditional probability reflects our belief which values of Y will occur more often or are more plausible given that we already know which values the rv X assumes.

If Y is independent from X:

$$\mathrm{P}(a \leq Y \leq b \,|\, x \leq X \leq x + \Delta x) = \mathrm{P}(a \leq Y \leq b)\,.$$



If (X, Y) have a joint density $p(x, y)$, the conditional density is

$$p_{Y|X}(y\,|\,x) = \frac{p(x, y)}{p_X(x)} \quad \text{for } p_X(x) \neq 0$$

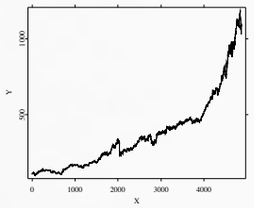
Then:

$$\mathrm{P}(a \leq Y \leq b\,|\,X = x) = \int_a^b p_{Y|X}(y\,|\,x)dy$$

and

$$\mathrm{E}[Y\,|\,X = x] = \int_{-\infty}^{\infty} y\, p_{Y|X}(y\,|\,x)dy = m(x)\,.$$

where $m(x) = \mathrm{E}[Y\,|\,X = x]$ is the conditional mean function of Y given $X = x$.



Theorem 3.1 Let $(X, Y, Z) \in \mathbb{R}^3$ *have a joint distribution.*

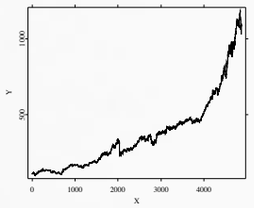
a) *If* X *and* Y *are independent:* $\mathsf{E}[Y \mid X = x] = \mathsf{E}[Y]$

b) *If* $Y = g(X)$:

$$\mathsf{E}[Y \mid X = x] = \mathsf{E}[g(X) \mid X = x] = g(x)$$

If $Y = Zg(X)$:

$$\mathsf{E}[Y \mid X = x] = \mathsf{E}[Zg(X) \mid X = x] = g(x)\mathsf{E}[Z \mid X = x]$$

c) $\mathsf{E}[aY + bZ \mid X = x] = a\mathsf{E}[Y \mid X = x] + b\mathsf{E}[Z \mid X = x]$

d) $\mathsf{E}_X\left[\mathsf{E}[Y \mid X]\right] = \mathsf{E}[Y]$